

Popularity Bias and Cold-Start in Patent Technology-Transfer Demand Prediction: A Diagnostic Evaluation of Graph- and Content-Based Link Prediction on KIPRIS

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Abstract

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Sangwon Lee: Investigation, Data curation, Visualization, Writing – original draft. **Kyungseok Lee:** Conceptualization, Methodology, Software, Formal analysis, Writing – original draft, Writing – review & editing. **Jaewon Lee:** Software, Validation, Investigation. **Dongkyu Lee:** Conceptualization, Supervision, Funding acquisition, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work

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Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used Claude (Anthropic) in order to improve the language, grammar, and readability of the manuscript. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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Data availability

The patent data were obtained from the Korea Intellectual Property Rights Information Service (KIPRIS) through its paid API (KIPRIS Plus, <https://plus.kipris.or.kr>) and were subsequently cleaned and processed by the authors. Because redistribution of these data is restricted by the KIPRIS terms of use, the processed dataset is not deposited in a public repository; it is instead available from the corresponding author by email upon reasonable request, subject to those terms. The preprocessing and evaluation code, experiment scripts, and an embedding-regeneration script are released at [[repository URL / DOI]], enabling full reproduction of all tables and figures.